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TRW Automotive

Slip Control Software Design

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1 ABSTRACT

The AyLimiter is designed to prevent static roll over, whereas Active Roll Mitigation ([ARM](#)) itself is designed to avoid dynamic roll over of the vehicle. This module contains all functionality of AyLimiter such as signal processing, activation, brake and engine controller.

2 ACRONYMS

ARM Active Roll Mitigation

ROI Roll Over Index

SWA Steering Wheel Angle

AyMod 'Modified' Lateral Acceleration, that tries to predict the Lateral Acceleration

Ay Lateral Acceleration

Thresh Threshold

Eng Engine

3 OVERVIEW

This document describes the purpose and behavior of Ay_Limiter.

3.1 DESIGN GOAL

The AyLimiter is designed to limit the lateral acceleration of the vehicle in steady state driving situations, that are characterized by:

- high lateral acceleration
- little dynamic
- small yaw rate raising
- small and slowly changes of steering angle

A prediction of the lateral acceleration A_{y_Lead} is calculated from the Lateral Acceleration (A_y) of the vehicle model due to driver steering angle $A_{y_From_Swa}$. The derivative of the signal is added as well, in order to predict the lateral acceleration in the soon future.

$A_{y_Limiter}$ is activated, when the predicted lateral acceleration and the measured lateral acceleration exceed particular thresholds.

A PID controller is used to react on sudden changes (derivative part) and ensure static control accuracy (integral part).

The intervention is performed in two stages. At first the engine torque is reduced. If this is not sufficient, a brake torque is requested at the wheels with the highest wheel load in order to decelerate the vehicle.

The PID controller consists of two proportional and two derivative controllers, whereas one considers the predicted lateral acceleration and the other considers the actual lateral acceleration. Thus the intervention is amplified very much, when the measured actual lateral acceleration exceeds the threshold. The control is performed at low slip levels with medium pressure build up rates. In case the maneuver evolve more dangerous ARM has to take over the control.

3.2 GENERAL REQUIREMENTS

1. $A_{y_Limiter}$ should prevent static roll over due to excessive cornering.

3.3 DETAILED REQUIREMENTS

Requirement ID	
Location	

3.4 SYSTEM MODE REQUIREMENTS

This module needs only to be processed, when ARM is allowed.

3.5 FLOW-CHARTS AND DIAGRAMS

None.

4 DETAILED DESIGN

This chapter outlines the detailed design of Arm_State .

4.1 ALGORITHM DESCRIPTION

4.1.1 AyLimiterMain

This function calls all sub-functions of the AyLimiter algorithm.

4.1.2 AyLimiterSignalProcessing

Steering Angle Derivative Filtered

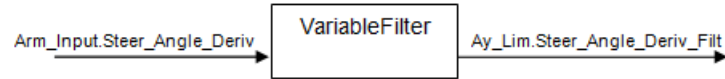


Figure 4.1: Steering Angle Derivative Filtered

Steering Angle Derivative Latched

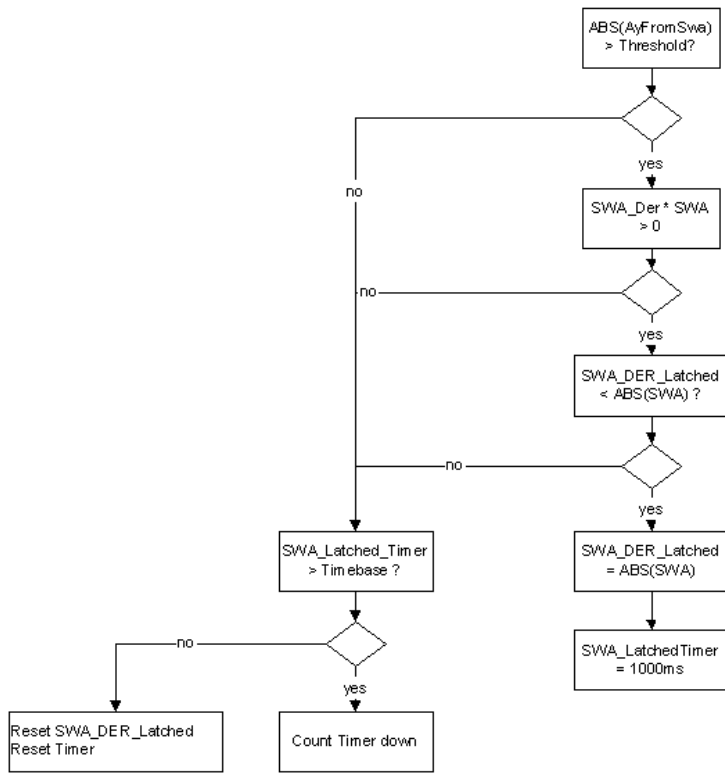


Figure 4.2: Steering Angle Derivative Latched

Ay From Steering Angle

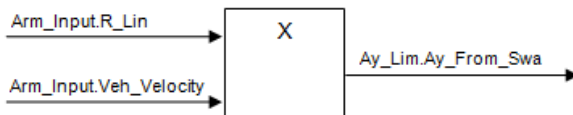


Figure 4.3: Ay From Steering Angle

Ay From Steering Angle Derivative

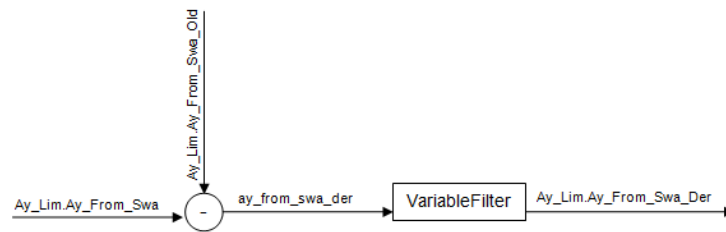


Figure 4.4: Ay From Steering Angle Derivative

Ay From Steering Angle Lead

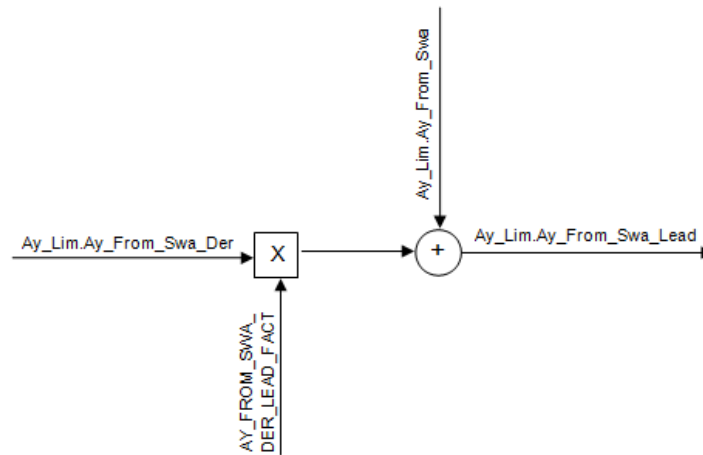


Figure 4.5: Ay From Steering Angle Lead

Ay_Lim.Lat_Accel_Roll_Corr_Filt

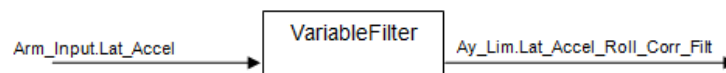


Figure 4.6: Ay_Lim.Lat_Accel_Roll_Corr_Filt

Ay_Lim_Lat_Accel_Der

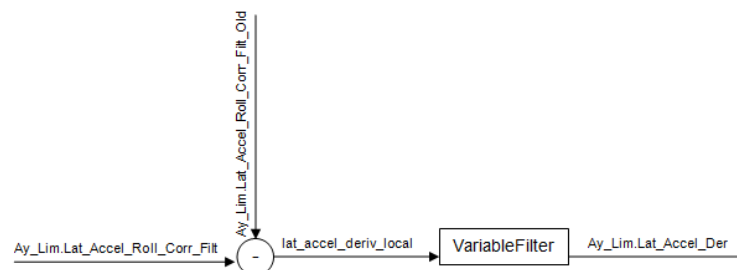


Figure 4.7: Ay_Lim_Lat_Accel_Der

4.1.3 AyLimiterVehicleStateCalc

This function calculates the necessity of an AyLimiter intervention. It takes several signal values into account to calculate whether an intervention by the AyLimiter should take place or not. The function calculates the activation and the deactivation of the AyLimiter.

The computation of activation and deactivation thresholds is split into several parts. The different thresholds are outlined in Fig 4.8. There are thresholds which are compared against lateral acceleration (red path) and those which are compared against a lateral acceleration estimated from the Steering Wheel Angle (SWA) (green path). The latter is called Ay_From_Swa.

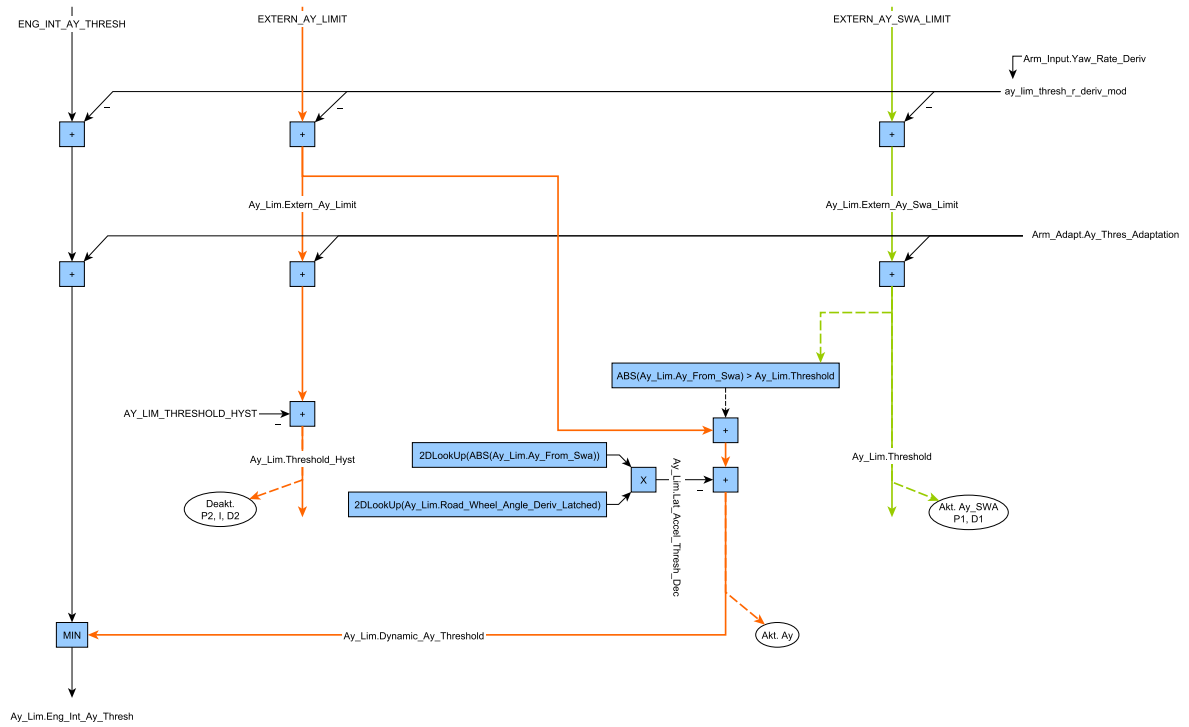


Figure 4.8: Computation of AyLimiter Thresholds

Basic Activation Thresholds

Basic Activation Thresholds are set by a Trim values that are reduced dynamically depending on yaw dynamics. One is for lateral acceleration and the other one is for Ay_From_Swa.

The reduction of AyLimiter activation thresholds due to yaw dynamics of the vehicle are a feature to extend the scope of AyLimiter from steady state maneuver towards more dynamic maneuvers, by sensitization of the activation thresholds. When yaw rate derivative exceeds a trim value, the activation thresholds are reduced by a trim value.

Basic Deactivation Threshold

Deactivation Threshold is set by Activation Threshold minus Hysteresis Trim Value.

Dynamic Activation Threshold

There is a second dynamic reduction which differs from the earlier yaw rate based reduction. To calculate the Dynamic Activation Threshold the Basic Activation Threshold is taken as basis. The basis threshold is reduced by a computed term, which is depending on the steering angle derivative latched value and the actual ay derived from steering angle.

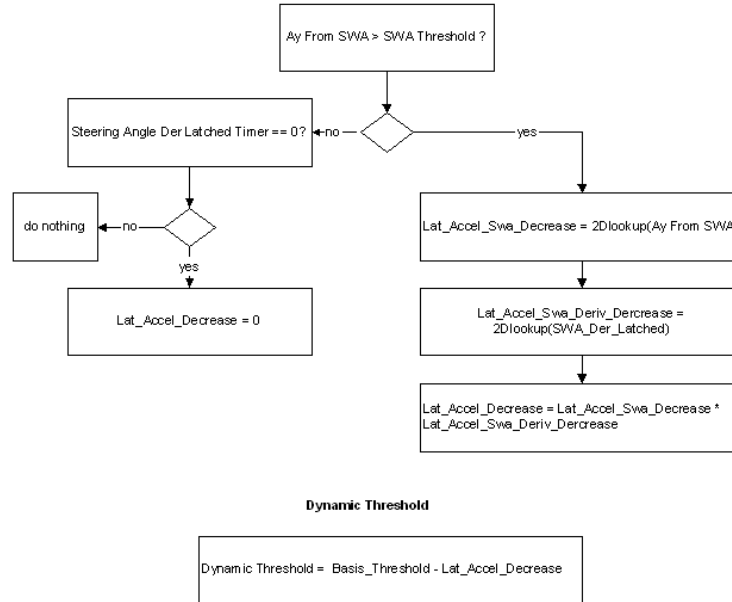


Figure 4.9: Dynamic Activation Threshold

Engine Activation Threshold

Engine Activation Threshold is set by a Trim value minus dynamic reduction due to yaw dynamics.

The reduction of AyLimiter activation thresholds due to yaw dynamics of the vehicle are a feature to extend the scope of AyLimiter from steady state maneuver towards more dynamic maneuvers, by sensitization of the activation thresholds. When yaw rate derivative exceeds a trim value, the activation thresholds are reduced by a trim value.

AyLimiter Reserve

Next step is to calculate the AyLimiter Reserve. This value can be used by other subsystems to evaluate the probability of an AyLimiter activation. Reserve values are in the interval [0 %, 100 %], where 100 % is equal to an AyLimiter activation. Note: The Reserve is calculated using the Basis Threshold and not the Dynamic Threshold. This is necessary to decrease oscillation of the signal in case of steering input by the driver.

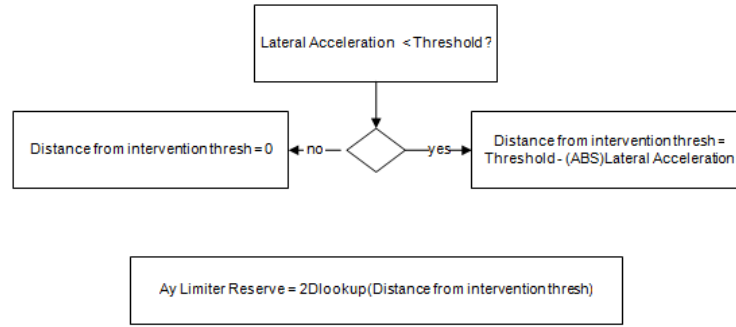


Figure 4.10: AyLimiter Reserve

Computation of Activation and Deactivation

Activation and deactivation of the ay limiter is computed as follows:

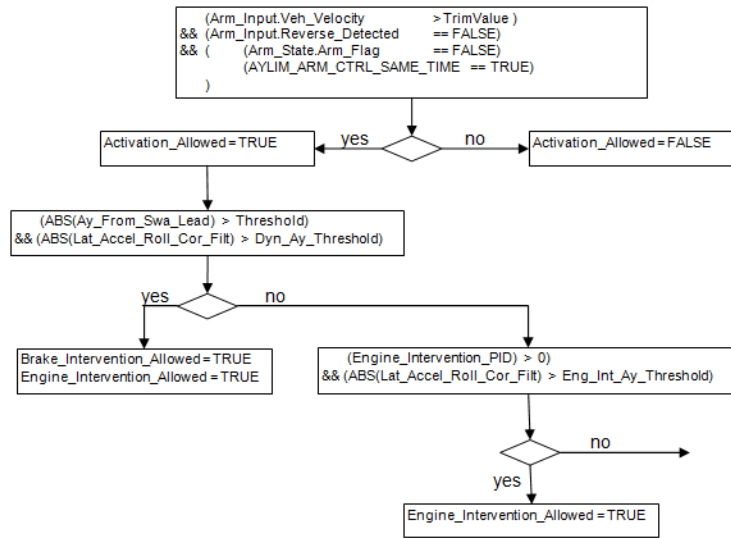


Figure 4.11: Computation of Activation and Deactivation

Offroad Mode

In offroad mode it is advised to disable the engine intervention. Therefore the computation in Fig. 4.11 is enhanced. In case the offroad switch is set and the brake temperature is normal, the engine intervention is forbidden for a trimable time (default: 2s).

4.1.4 AyLimiterPIDCalculation

This subsystem calculates the PID output of the Ay limiter algorithm, taking the control deviation, the derivative of the control deviation and the integrated value of the control deviation into account. The control deviation consist of two different parts:

- one part is the difference between lateral acceleration derived form steering angle and the intervention threshold,
- the second part is the difference between measured lateral acceleration and the intervention threshold.

In more detail the output of this function is a $P_1 D_1 I_1 P_2 D_2$ value.

The output of the PID controller is computed as follows:

AyLimiter Control Deviation P1

$$\text{Deviation}_1 = A_{y,\text{from_swa}} - \text{Treshold} \quad (4.1)$$

AyLimiter Control Deviation 2

$$\text{Deviation}_2 = A_{y,\text{meas}} - \text{Treshold_Hyst} \quad (4.2)$$

PID Calculation

The computation is outlined below in EQ. 4.3. Note, the derivatives are only calculated during steering increase.

$$\begin{aligned} PPIDD = & P_{1,\text{gain}} \cdot f_1(\text{Deviation}_1) \cdot \text{Deviation}_1 \\ & + P_{2,\text{gain}} \cdot \text{Deviation}_2 \\ & + D_{1,\text{gain}} \cdot f_2\left(\frac{d}{dt}\text{SWA}\right) \cdot \frac{d}{dt}A_{y,\text{from_swa}} \\ & + D_{2,\text{gain}} \cdot f_3(\text{Deviation}_2) \cdot \frac{d}{dt}A_{y,\text{meas}} \\ & + I_{\text{gain}} \cdot \int \text{Deviation}_2 \cdot dt \end{aligned} \quad (4.3)$$

4.1.5 AyLimiterEngineIntervention

This function computes the absolute value of the AyLimiter engine request. The AyLimiter Engine Request becomes active earlier to the AyLimiter brake intervention. As start point for control the engine torque is used, that was actual, when the predicted lateral acceleration due to steering input exceeds the threshold.

After control the actual engine torque (which should be equal to the AyLimiter engine request) is ramped towards the driver requested engine torque.

At the start of control the maximum amount of the PID output, which is used for engine control is calculated:

$$\text{PIDEngine} = \text{LIMIT}(\text{PID}, 0, \text{MAX_ENGINE_INTERVENTION}) \quad (4.4)$$

Engine Intervention is activated the following conditions are TRUE:

- Arm_Engine_Enabled
- Engine_Intervention_Allowed
- Driver_Requested_Torque > 0

If the engine intervention is activated the absolute value of requested engine torque is computed as follows:

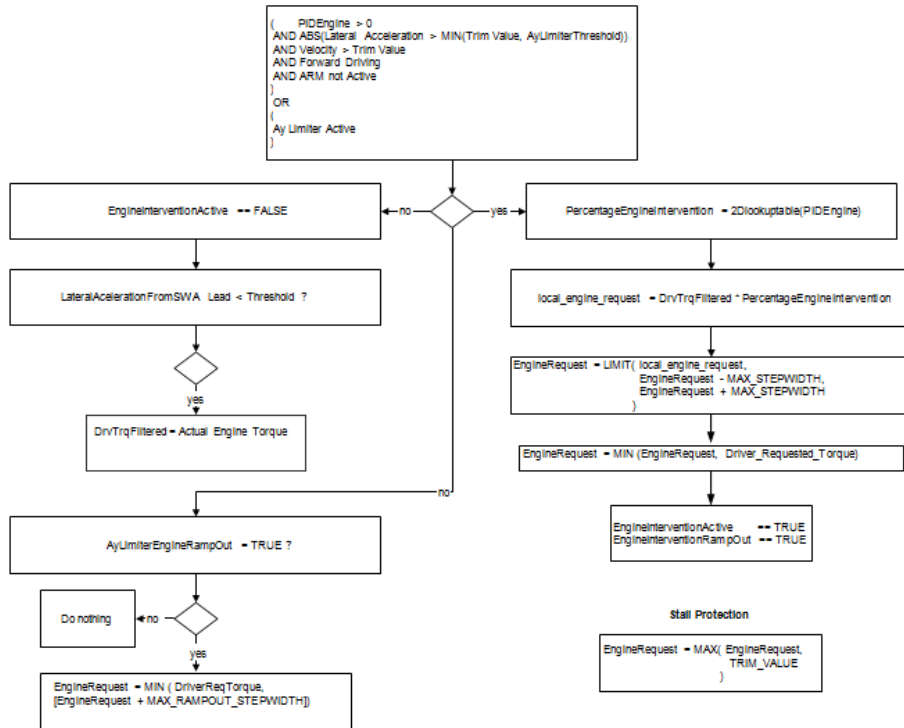


Figure 4.12: Computation of Engine Intervention

Too avoid too harsh engine acceleration if the driver is requesting higher torque than the AyLimiter right after AyLimiter deactivation, the engine request is ramped towards the driver requested engine torque request:

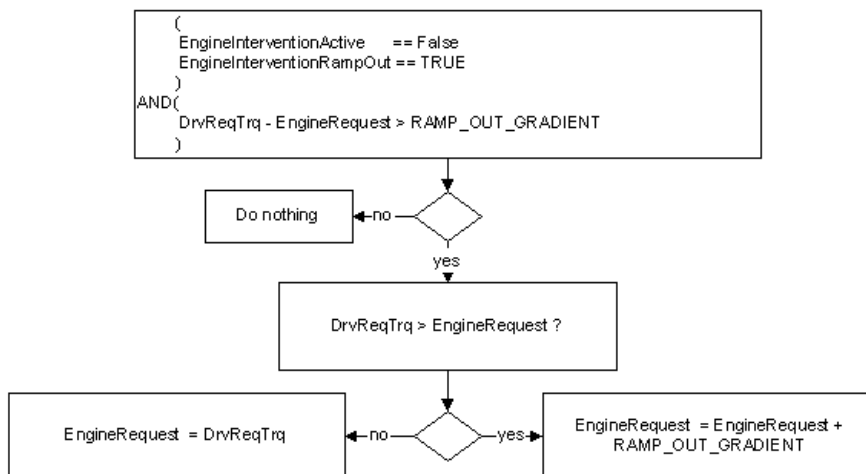


Figure 4.13: Ramp Out of Engine Intervention

4.1.6 AyLimiterBrakeIntervention

This function computes the AyLimiter Brake Pressure, which is applied to the curve outer front and rear wheels.

AyLimiter Brake intervention is computed using the output of the PID controller.

The PID Output is decreased by the PID Part which was transferred to the engine intervention:

$$PID_{BrakeIntervention} = PID - PID_{Engine} \quad (4.5)$$

Brake intervention is only activated, if

1. AyLimiter brake intervention is allowed and
2. $PID > PID_{Engine}$

The Raw brake intervention is computed as follows:

$$BrakeInterventionRawValue = Gain \cdot PID_{BrakeIntervention} \quad (4.6)$$

The rise rate is limited by the brake torque gradient calculation:

$$BrakeInterventionPID = MIN(BrakeInterventionRawValue, BrakeInterventionPID + loop_delta) \quad (4.7)$$

with

$$loop_delta = f(BrakeTorqueGradient) \quad (4.8)$$

The Pressure Request is limited with a maximum value:

$$BrakePressurePrimWheel = LIMIT(BrakeInterventionPID, 0, PRIM_MAX_VALUE) \quad (4.9)$$

$$BrakePressureSECCWheel = LIMIT(BrakeInterSec, 0, SEC_MAX_VALUE) \quad (4.10)$$

with

$$BrakeInterSec = BrakeInterventionPID \cdot BRAKE_PRESSUER_SEC_GAIN \quad (4.11)$$

Pressure Release while the AyLimiter is active is limited to avoid to harsh pressure releases:

$$BrakeInterventionPIDPrim = MAX(BrakePressurePrimWheel, \quad (4.12)$$

$$BrakeInterventionPIDPrim - Stepwidth) \quad (4.13)$$

$$BrakeInterventionPIDSec = MAX(BrakePressureSECCWheel, \quad (4.14)$$

$$BrakeInterventionPIDSec - Stepwidth) \quad (4.15)$$

To avoid a too harsh pressure release if the AyLimiter is deactivated the pressure release rate is limited:

$$BrakeInterventionPIDPrim = MAX(BrakeInterventionPIDPrim - PRESS_RAMP_OUT_STEPWIDTH, 0) \quad (4.16)$$

$$BrakeInterventionPIDSec = MAX(BrakeInterventionPIDSec - PRESS_RAMP_OUT_STEPWIDTH, 0) \quad (4.17)$$

$$(4.18)$$

A maximum Value of brake slip is set:

$$MaxAllowedBrakeSlip = 10\% \quad (4.19)$$

4.1.7 AyLimiterBrakeTorqueGradient

This function computes the pressure rise rate requested by the AyLimiter algorithm.

This function calculates the pressure gradient requested by the AyLimiter. The separate request is necessary, as the current implementation of the rise rate calculation within the YSC subsystem is not sufficient for the AyLimiter specification.

The current implementation in the YSC rates low pressure amplitudes with low gradients but the AyLimiter requests high pressure gradients in dynamic maneuvers together with low pressure amplitudes.

The calculation is done as follows:

$$\text{brake_torque_gradient_gain} = 2\text{Dlookup}(\text{SWA_Derivative_Latched}) \quad (4.20)$$

The gradient is calculated for each wheel individually.

4.1.8 AyLimiterOutput

This function allocates all output variables of the AyLimiter algorithm. In case the AyLimiter is switched of by compile switch AY_LIMITER, the function AyLimiterOutput returns default values.

5 PROVIDED TASKS

5.1 AyLimiterMain

Description	AyLimiterMain
Rate	Standard Arm timebase 5 - 20ms
Dependency	
SYNTAX	void AyLimiterMain (void)

5.2 AyLimiterOutput

Description	AyLimiterOutput
Rate	Standard Arm timebase 5 - 20ms
Dependency	
SYNTAX	void AyLimiterOutput (void)

6 PROVIDED SERVICES

6.0.1 Private Services

- [AyLimiterSignalProcessing](#)
- [AyLimiterVehicleStateCalc](#)
- [AyLimiterPIDCalculation](#)
- [AyLimiterEngineIntervention](#)
- [AyLimiterBrakeIntervention](#)
- [AyLimiterBrakeTorqueGradient](#)

7 CONFIGURATION OPTIONS

None.

8 CONFIGURATION PROCEDURES

None.

9 TEST CASES

Test category	
Requirements	
Test Case ID	

10 CALIBRATION INFORMATION

Like [ARM](#), the behavior of AyLimiter can be adjusted by parameters. These different adjustments are outlined in the following sections. In General, AyLimiter needs to be optimized for steady cornering, continuous steering (performance) and slalom (robustness).

10.1 Activation

The activation is split into a threshold for steering and a threshold for measured lateral acceleration. In general, there are two global offsets to sensitize AyLimiter. An detailed overview of thresholds can be found in [ARM-DDD-Ay_Limiter.pdf](#).

- **AY_LIM_THRESH_R_DERIV_MOD**
- **Arm_Adapt.Ay_Thres_Adaptation**

These offsets are applied to every AyLimiter threshold.

10.1.1 Steering

The parameter to make AyLimiter more sensitive on steering only is **EXTERN_AY_SWA_LIMIT**. By lowering this threshold, AyLimiter will activate earlier and the intervention will be stronger since the steering control error is computed based on this threshold. In Addition, the dynamic lateral acceleration threshold is more sensitive if **Ay_Lim.Threshold** is low.

10.1.2 Lateral Acceleration

The activation due to a high lateral acceleration, is adjustable through **EXTERN_AY_LIMIT**. The control error is computed based on

$$\text{EXTERN_AY_LIMIT} - r_deriv_mod - \text{AY_LIM_THRESHOLD_HYST} \quad (10.1)$$

and measured lateral acceleration.

Furthermore the dynamic reduction can be used to activate earlier depending on **Ay_From_Swa** and **RWA_Deriv_Latched**. The reduction will be computed if the steering threshold is already exceeded, but the lateral acceleration is still a bit too low for activating AyLimiter. Use

- **AY_LIM_THR_DEC_AY_SWA_TBL** for offset
- **AY_LIM_THRESHOLD_DEC_TBL** for scaling the offset

10.2 Deactivation

The basic deactivation threshold is outlined in Eq. 10.1. The parameter **AY_LIM_THRESHOLD_HYST** is used to avoid toggling and in/out control.

10.3 Engine Intervention

The engine intervention can be adjusted by changing **MAX_AMOUNT_ENGINE_INTERVENTION**. This parameter will influence the brake intervention since the PID-Sum will be reduced by this parameter. The intervention strength is controlled by **ENGINE_REDUCTION_TBL** and the general PID parameters.

10.4 Brake Intervention

Brake and engine intervention are strongly connected. It is worth adjusting the PID parameters.

10.5 AyLimiter Reserve

AyLimiter Reserve can be used to fade out the **YSC!** understeer intervention. Currently the fade out is not correctly implemented in **YSC!**. The distance from lateral acceleration to the dynamic activation threshold is used to determine how certain it is that AyLimiter will activate in the future. One can use **AY_LIM_RESERVE_TABLE** to adjust the behavior and the understeer fade out. The table maps the physical distance (acceleration) to a percentage that is directly multiplied with the understeer pressure request.

11 APPENDIX